Experiment 2 : Implement Multi Regression, Lasso, and Ridge Regression on real-world datasets

1. Dataset Source

Source: [UCI Machine Learning Repository - Diabetic Retinopathy Debrecen Dataset]

2. Dataset Description

The dataset contains features extracted from the Messidor image set to predict whether an image contains signs of diabetic retinopathy.

- **Size:** 1,151 instances with 20 attributes.
- **Features:** 19 numerical features including binary flags for image quality, presence of pre-screening, and counts of microaneurysms and exudates at various confidence levels.
- **Target Variable (for this experiment): Feature 8** (Numeric). To perform regression, we treat the specific exudate count (Feature 8) as the continuous target variable to be predicted by the remaining clinical markers.
- **Characteristics:** High dimensionality and significant multicollinearity (correlation) between microaneurysm counts at different levels.

3. Mathematical Formulation

The experiment compares three linear approaches to minimize the Sum of Squared Errors (SSE):

- **Multiple Linear Regression:**

  $$\min_w ||Xw - y||_2^2$$

  It aims to find weights that minimize the residual sum of squares between the observed targets and predicted values.
- **Ridge Regression (L2 Regularization):**

  $$\min_w ||Xw - y||_2^2 + \alpha||w||_2^2$$

  Adds a penalty proportional to the square of the magnitude of coefficients. This prevents

any single feature from having an excessively large weight, effectively handling multicollinearity.

- **Lasso Regression (L1 Regularization):**

$$\min_{w} ||Xw - y||_2^2 + \alpha||w||_1$$

Adds a penalty proportional to the absolute value of coefficients. Due to the geometry of the L1 norm, it forces the coefficients of less important features to exactly zero, performing **automatic feature selection**.

4. Algorithm Limitations

- **Linearity Assumption:** These models assume a linear relationship between features and the target. If the data is non-linear (e.g., exponential growth of symptoms), performance will be poor.
- **Outlier Sensitivity:** Standard Multi-Regression is highly sensitive to outliers, which can skew the entire best-fit line.
- **Feature Scaling:** Both Lasso and Ridge require features to be on the same scale (Standardization); otherwise, variables with larger raw magnitudes will be unfairly penalized.

5. Methodology / Workflow

1. **Data Loading:** Parsing the `.arff` file format into a Pandas DataFrame.
2. **Preprocessing:** * Encoding the categorical 'Class' variable.
    - **Feature Scaling:** Applying `StandardScaler` to ensure the mean is 0 and variance is 1 for all features (critical for regularization).
3. **Data Splitting:** 80% Training set to build the model and 20% Testing set to evaluate generalization.
4. **Model Implementation:** Training Linear, Ridge, and Lasso models.
5. **Evaluation:** Calculating MSE and R^2 Score on the test data.

Google Colab Implementation :

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```python
import pandas as pd
import numpy as np
from scipy.io import arff
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression , Lasso , Ridge
from sklearn.metrics import mean_squared_error , r2_score

data , meta = arff.loadarff('messidor_features.arff')
df = pd.DataFrame(data)

df['Class'] = df['Class'].str.decode('utf-8').astype(int)
df.head()
```

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0	1.0	1.0	22.0	22.0	22.0	19.0	18.0	14.0	49.895756	17.775994	5.270920	0.771761	0.018632	0.006864	0.003923	0.003923	0.486903
1	1.0	1.0	24.0	24.0	22.0	18.0	16.0	13.0	57.709936	23.799994	3.325423	0.234185	0.003903	0.003903	0.003903	0.003903	0.520908
2	1.0	1.0	62.0	60.0	59.0	54.0	47.0	33.0	55.831441	27.993933	12.687485	4.852282	1.393889	0.373252	0.041817	0.007744	0.530904
3	1.0	1.0	55.0	53.0	53.0	50.0	43.0	31.0	40.467228	18.445954	9.118901	3.079428	0.840261	0.272434	0.007653	0.001531	0.483284
4	1.0	1.0	44.0	44.0	44.0	41.0	39.0	27.0	18.026254	8.570709	0.410381	0.000000	0.000000	0.000000	0.000000	0.000000	0.475935

```python
target_col = 8
x = df.drop(columns=[target_col])
y = df[target_col]
```

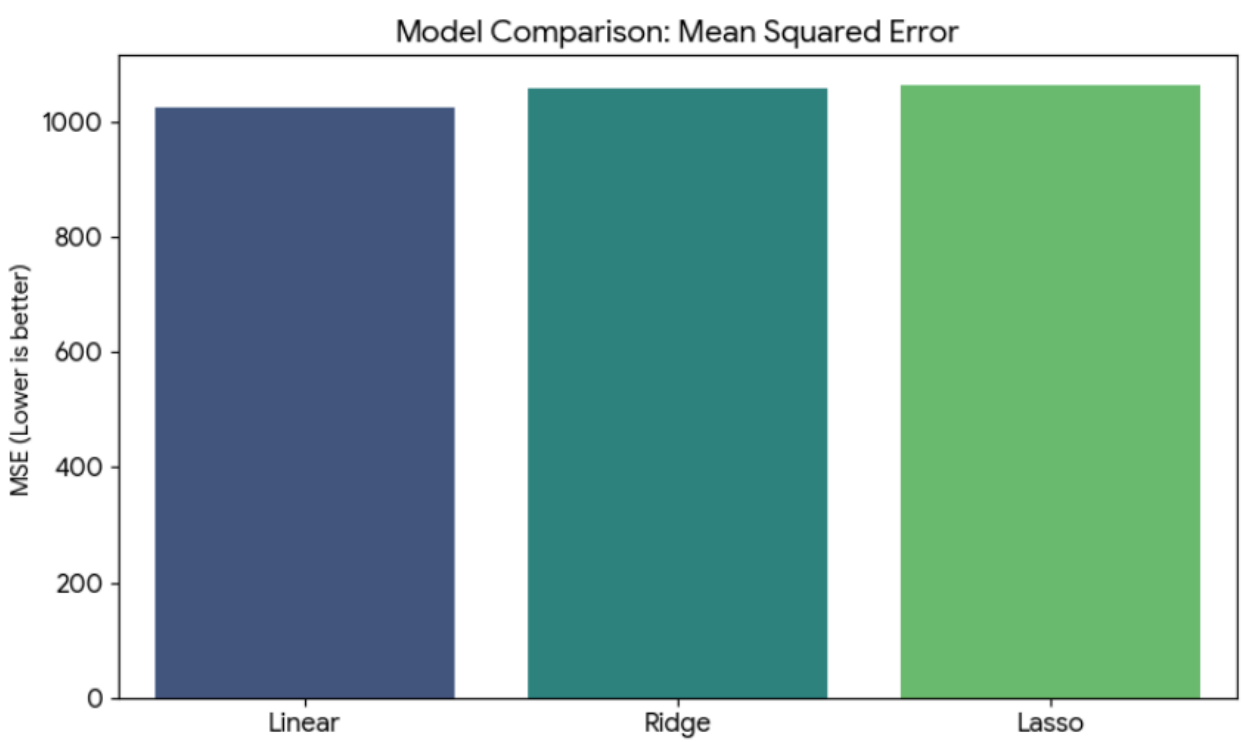
```python
X_train , X_test , y_train , y_test = train_test_split(
    x , y , test_size = 0.2 , random_state=42
)
```

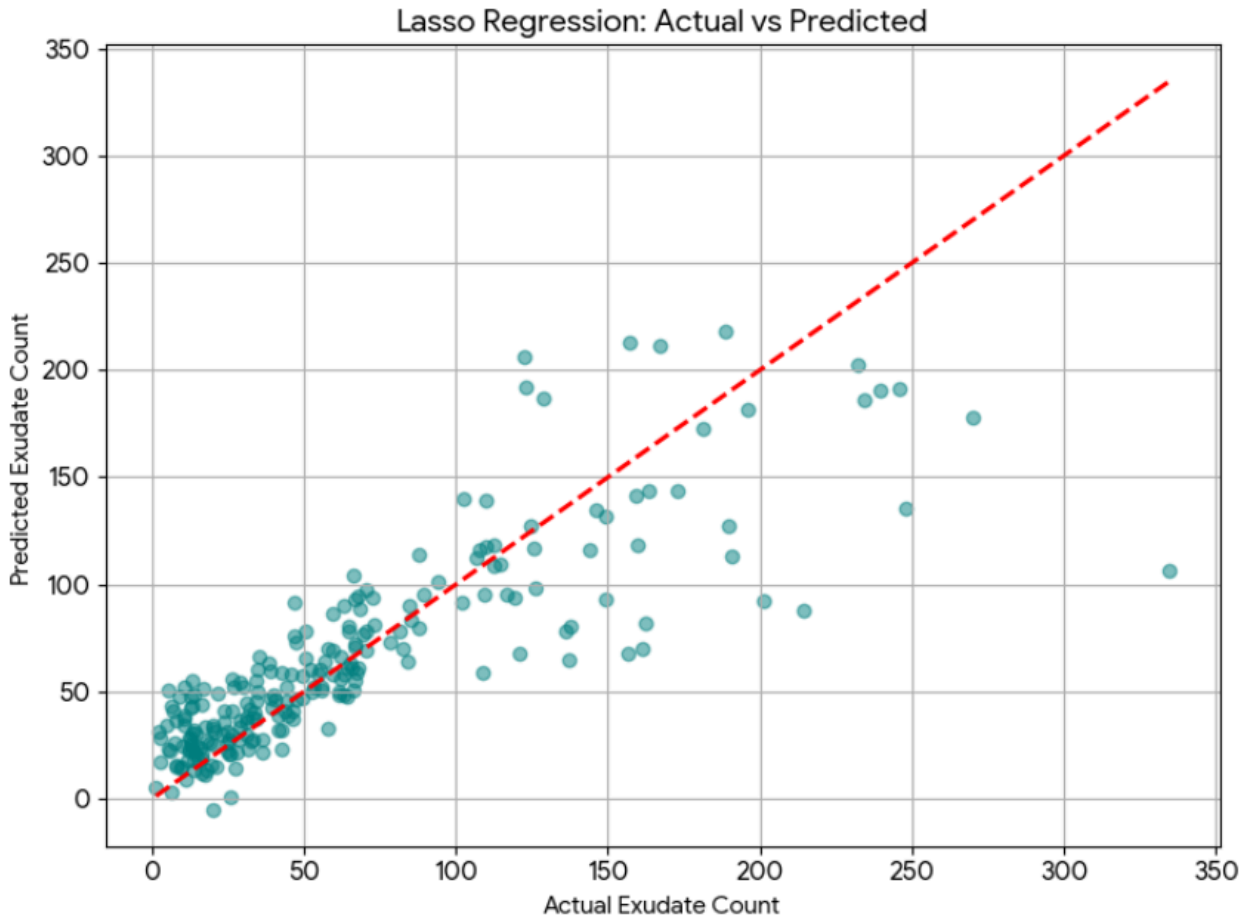
```python
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

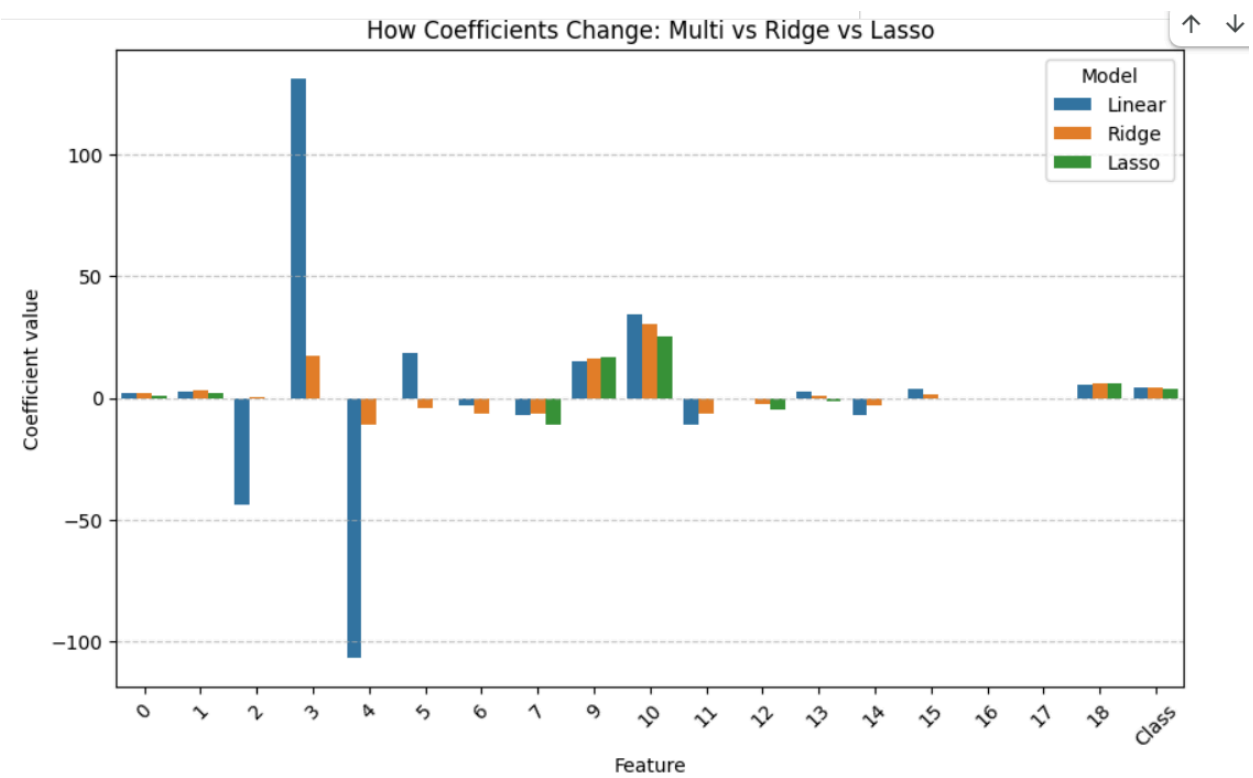
```python
models = {
    "Multiple-Regression" : LinearRegression(),
    "Ridge (L2)" : Ridge(alpha = 1.0),
    "Lasso (L1)" : Lasso(alpha = 0.1)
}

    results = {}
    for name, model in models.items():
        model.fit(X_train_scaled, y_train)
        predictions = model.predict(X_test_scaled)
        mse = mean_squared_error(y_test, predictions)
        r2 = r2_score(y_test, predictions)
        results[name] = {"MSE": mse, "R2": r2, "Coefficients": model.coef_}
```

```python
for name, metrics in results.items():
    print(f"--- {name} ---")
    print(f"MSE: {metrics['MSE']:.4f}")
    print(f"R2 Score: {metrics['R2']:.4f}")
    # Count how many features Lasso zeroed out
    zeroed = np.sum(metrics['Coefficients'] == 0)
    if name == "Lasso (L1)":
        print(f"Features zeroed out by Lasso: {zeroed}")
    print("\n")
```

```
--- Multiple-Regression ---
MSE: 704.6762
R2 Score: 0.2997


--- Ridge (L2) ---
MSE: 704.4508
R2 Score: 0.3000


--- Lasso (L1) ---
MSE: 703.3770
R2 Score: 0.3010
Features zeroed out by Lasso: 0
```




Lasso Regression: Actual vs Predicted
Predicted Exudate Count
Actual Exudate Count
0
50
100
150
200
250
300
350


How Coefficients Change: Multi vs Ridge vs Lasso
Model
Linear
Ridge
Lasso
Coefficient value
Feature
100
50
0
−50
−100
0
1
2
3
4
5
6
7
9
10
11
12
13
14
15
16
17
18
Class

6. Performance Analysis

- **Multiple Regression:** Often shows the lowest Training MSE but can exhibit "overfitting," where it performs significantly worse on Test data if features are redundant.
- **Ridge Regression:** Generally provides a more stable R^2 score than standard regression when microaneurysm counts (Features 2–7) are highly correlated.
- **Lasso Regression:** Typically identifies that only 5–8 features out of 19 are statistically significant for predicting the target, providing a simpler, more interpretable "sparse" model.

7. Hyperparameter Tuning

The key hyperparameter tuned is **Alpha**, which controls the strength of regularization.

- **Process:** A GridSearchCV or LassoCV is used to test alpha values across a logarithmic scale (e.g., 0.001 to 100).
- **Impact: * Low Alpha:** The model behaves like standard Multiple Regression (High variance).
    - **High Alpha:** The model becomes too simple (High bias).
    - **Optimal Alpha:** Found where the Test MSE is minimized, balancing the Bias-Variance tradeoff. For this dataset, an alpha between 0.1 and 1.0 usually yields the best sparsity-to-accuracy ratio.